ME 315
HEAT AND MASS TRANSFER

CRN: 33797
Spring 2026 Syllabus

Lecture:

Division 002: MWF 8:30-9:20 AM, ME G061
Professor Thomas Beechem (email: tbeechem@purdue.edu)

Division 001: MWF 9:30-10:20 AM, ME G061
Professor Justin Weibel (email: jaweibel@purdue.edu)

Division 049: MWF 3:30-4:20 PM, BRNG 2290
Mr. Sudharshan Anandan (email: sanandan@purdue.edu)
2026 Lambert Fellow

Course TA
Ziqi Fang (fang212@purdue.edu)

Instructional Modality: Face-to-Face

Recitation: **Videos Posted Online.** The recitation solves example problems to help understanding and exam preparation. It is entirely asynchronous even as your schedule may indicate a time and location.

Instructional Modality: Async-Online

Laboratory: **ME315L Heat and Mass Transfer Laboratory**
Professor Xianfan Xu, Lab Supervisor

Lab information including assignments and other policies will be transmitted by your lab instructor.

Class Website: **Brightspace & Gradescope.** Details provided below.

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Course Objectives

ME 315 serves to introduce the student to the many different processes by which energy and mass may be transferred. A list of the specific topics is provided in the attached Assignment Schedule. The course pursues three overriding objectives:

(a) The student should appreciate the physical origins of the various transport mechanisms, including conduction, radiation, and convection. Moreover, when confronted with a particular problem, the student should be able to identify the relevant transport processes.

(b) The student should be able to perform engineering calculations for problems involving heat and mass transfer. The student should know when simplifying approximations may be made and the impact of those assumptions.

(c) The student should be able to perform calculations and apply assumptions that lead to a rational design and/or an improved understanding of the performance of thermal and mass exchange systems.

(d) A final, yet equally important objective, is to develop a positive attitude towards the subject of heat and mass transfer. It is incumbent upon the instructor to reveal the vital role that such processes play in the natural and industrial worlds and to thereby transmit a sense of excitement for the subject.

Ultimately, the student should leave the course with confidence, understanding of, and the ability to apply the basic principles of heat transfer. You should also leave with a desire to apply what you have learned and to expand upon your background in thermo- and fluid dynamics.

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1. ***Prerequisites.*** ME 315 must be preceded by ME 308. If you do not meet this requirement, see your instructor immediately.

2. ***Class Websites***.
   - Each division (lecture section) will have a unique Brightspace page for individual lecturers to post notes, announcements, and other division specific information. You will have access to the division that you are enrolled. Hereafter, the Brightspace page particular to your division will be termed the "Division Brightspace page."
   - All ME315 students regardless of division will also have access to the "Total Course Brightspace page." This ME315-wide Brightspace page will be used to post items relevant to all divisions. This includes homework solutions, lab materials, recitation videos, along with exam preparation and review materials, etc. Hereafter, this ME315-wide Brightspace page will be termed the "Course-wide Brightspace page."
   - Homework assignment posting and submission for the lecture portion of the class will take place via ME315 Heat and Mass Transfer Gradescope page. Lab related assignments (prelabs, reports, presentation, etc.) will also be submitted to this Gradescope page according to lab-section-specific due dates. Gradescope will also be used to post the tabulation of marks. The Gradescope page link is here, and you will be added to the site by instructors soon after class starts (Monday, January 12, 2026). If you are enrolled after the start date or have troubles to access the page, you need to email the lead TA, Ziqi Fang (email above) for support.

3. ***Lecture.*** You are expected to participate in each lecture in the following manner:
   - The lecture only highlights materials, concepts, and works a small number of illustrative problems. The lecture is not exhaustive. You ***must*** do the reading corresponding to a particular lecture in advance. The reading assignments are posted with their corresponding lecture within the "Reading" column of the Class and Assignment Schedule below.
   - The instructors and course TAs will hold regular office hours in-person. Please see the office hour schedule posted on the Course-wide Brightspace page. We encourage participation in office hours.
   - **Attendance.** Classroom attendance is required except in cases of illness, emergencies, or other special circumstances as denoted by the university policies (see this Link). In case of an absence, an email notice should be sent to the lecturer before the lecture or as soon as you are able. You will be held responsible for all material discussed in lecture, whether treated in the assigned text or not. There will be no opportunity for make-up of missed assignments in the daily work (homework and quizzes/in-class exercises), except in cases of excused absences processed by the Office of the Dean of Students (ODOS).

- **Participation**. You are encouraged to ask questions during the lecture regarding aspects of reading, homework, or lecture material that is unclear to you. In addition, you may be called upon to answer questions, to comment on problem solutions, and/or lead discussions related to the lecture material. Demonstrating reasonable participation will require daily preparation and staying current with the assignments.

4. ***Quizzes and In Class Exercises*.** Short quizzes or in class exercises, typically of duration 5 to 10 minutes, will be given periodically in the lecture, generally without prior announcement, to highlight key points and to promote individual participation in the lecture. These quizzes may involve material from reading or homework assigned for the lecture or material developed during the lecture. Remember to bring a calculator to each lecture.

5. ***Homework***: Homework problems will be posted on the ME315 Heat and Mass Transfer Gradescope page and will be assigned approximately weekly and due *by 11:59 pm on the due dates shown in the Class and Assignment Schedule*. The due date and time will also be denoted in Gradescope.

   No late homework will be accepted.

   Gradescope enables online grading and viewing of graded assignments. Details of how to upload your homework including a video and a PDF guide for students are available at this Link. The Course-wide Brightspace page provides a link to Gradescope. Use that link to submit your homework. If the website is not accessible on the due date, homework may be emailed to the course TA (Ziqi Fang, email provided above) by the assigned due date. The subject line should read "ME315-Homework #-First Name-Last Name". In the case that an attached file emailed to the TA cannot be opened (i.e., it is corrupted), the assignment will be considered late and not accepted.

   Part of your submission is assigning your work to each question according to the assignment outline on Gradescope. Failure to assign your work to individual questions in Gradescope will result in a grade penalty. If submitting close to the assignment deadline, recognize that assigning questions is a requirement and budget time for this task accordingly.

   All homework must be completed using handwritten solutions. See the final page of this document for a template outlining how homework should be formatted. This template should be used as a guide in solving all your assigned homework problems. It is not necessary to include sections from the template if they are not relevant to your solution.

   Homework assignments will be evaluated primarily on completion and demonstration of a reasonable level of accuracy, consistent with the instructional aims of each problem. All problem solutions will be posted on the Course-wide Brightspace page for your further study.

   Assigned the week before exams or quiet week, homework assignments #5, #9, and #12 will **not** be graded (see schedule below). While not graded, the material included in these assignments **will be** "fair game" for inclusion on the relevant exam(s). Completion of all assignments, graded or otherwise, is **strongly encouraged** to perform well on the exams.

6. ***Exams***. Two midterm exams and one final exam will be given on the scheduled dates and will cover the designated material. No make-up exams will be given.

Students with excusable absences (per university policy: grief/bereavement, military service, jury duty, parenting leave, or medical excuses) will be assigned a grade equal to the average of their class-weighted grade of the other exams. Consider, for example, a student that places in the top 20% and 30% of all students on Exam #1 and the final, respectively, but misses Exam #2 due to an excused absence. They would be assigned a grade on Exam #2 that was in the top ~26% for that exam, where the higher weight of the final has been taken into account. Excused absences are **only** granted through the Office of the Dean of Students (ODOS) website.

Students must follow the university policies and contact the Office of the Dean of Students to obtain permission for an excusable absence. Related to medical issues, the policy states:

> *Students may occasionally miss class and other academic obligations due to emergent, serious, short-term, health-related reasons, whether physical or mental. These would include visits to an emergency room, hospitalization, or surgery. The Medical Excused Absence Policy intends to afford arrangements for students experiencing these medical situations who, at the direction of the medical provider, missed coursework or exams.*

If the final exam is missed (without a legitimate excuse noted above), students will be assigned a "failed (F)" grade. In the event of an excusable absence, the student will be assigned an incomplete grade. They will be able to take the ME315 final in a subsequent semester for course completion apart from having to retake the class as a whole.

Exams will be closed book in nature. Students will be provided an “Equation Sheet” on the exam containing relevant relations needed for problem solving. An example equation sheet will be posted to the Course-wide Brightspace page approximately 2-weeks prior to the exam. This minimizes the need for memorization thereby allowing students to focus on analysis techniques. No additional resources (i.e., crib-sheets, textbook etc.) will be available during the exams.

7. ***Recitation*.** Starting from the second week of the semester, recitation videos will be posted weekly to the Course-wide Brightspace page each week. These recitation videos aim to discuss and solve example problems to help students' understanding and exam preparation. Watching the recitation videos is optional but highly recommended for all students.

8. ***Grades.*** The course grade will be based on homework, quiz, and exam results along with the laboratory grade, which will be forwarded by your Laboratory Instructor to the Lecturer. The following list gives the weighting of the various items to be used in the determination of grades for the course:

Homework Problems	10%
Quizzes/In-Class Exercises	5%
Midterms (2)	35%
Final Examination	25%
Laboratory	25%
Total	100%

Final grade assignments will follow a standard scale:

- A (including A+ and A-): 90-100%
- B (including B+ and B-): 80-89.99%
- C (including C+ and C-): 70-79.99%
- D (including D+ and D-): 60-60.99%
- F: 0-59.99%

At the instructors' discretion, the final grades may be curved considering the overall performance of all students in the course. Note that curving will only improve a student's letter grade compared to the standard scale given above.

A passing grade in the course cannot be given without satisfactory performance in the laboratory component culminating in at least a passing laboratory grade more than 59.99%. (See the Laboratory Policy & Procedures in the Lab section of Brightspace for the definition of "Satisfactory Laboratory Performance".)

9. ***ME Exam Calculator Policy.*** The School of Mechanical Engineering's calculator policy **will be** enforced. The policy can be found at this Link, where the acceptable models of calculators permitted for exam use are outlined. Students retain the freedom to use their calculator of choice on homework, labs and projects, but are encouraged to practice using their exam calculator to make sure they fully understand its functionality.

Students who violate this policy are subject to sanctions from the instructor, up to and including a failing grade on the exam or even a failing grade in the course. Furthermore, all infractions may also be reported to the Office of the Dean of Students as potential violations of the Purdue University Code of Student Conduct which may result in additional sanctions being assigned, including probation, suspension, or even expulsion from the University. Any questions about this policy should be addressed to your instructor.

10. ***University Policy Statements***

(a) ***Unethical Conduct.*** Communication between students in solving homework problems and in working lab assignments is encouraged. However, all students are expected to do their own work in satisfying the homework and lab requirements, without using any

materials that are not conducive to meeting learning objectives and are not provided by instructors to this end. This includes – but is not limited to - solutions posted online, past homework assignments, and any actions in which another's work is represented to be that of one's own. Plagiarizing (copying) part, or all, of any assignment (homework, pre-lab, and/or lab report) from another student, from previous semesters, from online sources, or using data not collected by your lab group will result in the student being reported to the Office of Student Rights and Responsibilities (OSRR) for review at the university level. A grade penalty will be assessed of up to and including a failing grade in the class. The penalty equally applies to students who allow others to copy their assignments and, on assignments submitted as a team. It is the responsibility of each team member to ensure the entire assignment is free of plagiarism. Any student detected of cheating on an examination will receive a failing grade in the course, and documentation will be sent to the Dean of Students Office, recommending termination from the University for a period of no less than one year.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

The Purdue Honor Pledge Force, a student organization responsible for stewarding the mission of the Honor Pledge and encouraging a culture of academic integrity, asks all instructors to prominently include the student-initiated Purdue Honor Pledge on their syllabus, as well as exams and key assignments:

> "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue"

(b) ***Commercial Note Taking in Classes***. Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in Purdue University courses are permitted to take notes, which they may use for individual/ group study or for other non-commercial purposes. Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor.

(c) ***Emergency Procedure***. In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Consult your course instructor for information about changes.

(d) ***Mental Health-CAPS Information.*** Purdue University is committed to advancing the mental health and wellbeing of its students.

- **If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack.** Sign in and find information and tools at your fingertips, available to you at any time.
- **If you need support and information about options and resources**, please see the Office of the Dean of Students for drop-in hours (M-F, 8 am- 5 pm).
- **If you're struggling and need mental health services**: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

(e) ***Nondiscrimination Statement.*** Link to Purdue's nondiscrimination policy statement.

(f) ***Accessibility and Accommodations Syllabus Statement.*** Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let the instructor know and to discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

11. ***Responsible & Ethical Use of Artificial Intelligence (AI) in ME315***

Advancements in AI are providing new channels of access to information and assistance with problem solving and other tasks. You are encouraged to develop practices in which you can use AI tools as an assistant to enhance your learning and understanding but should not rely on AI to complete work on your behalf. Avoid reliance on AI to complete tasks in a manner that bypasses your need to understand the course material or develop the skills necessary to be successful in the course assessments (you will not have access to these tools during exams). You are ultimately responsible for all of your own work; carefully consider the output of all AI-generated content, and be aware AI is known fabricate false information.

Expectations and guidelines for the ethical and responsible use of AI in ME315.

- **Original Work:** Students should ensure that assignments submitted are original and based on their understanding. While AI can assist in research or provide general guidance, it should not create or complete work on behalf of the student.
- **Collaboration:** If a student collaborates with AI tools, they must specify the nature and extent of this collaboration in their submission. This includes providing details of the prompts used to generate the AI responses. *This step is considered part of a complete homework submission.*
- **Citation:** Any content, ideas, or assistance obtained through AI tools must be appropriately cited, like any other reference or source. You must find the relevant *original* source of information (journal articles, website, books) and cite this source in addition to acknowledging use of the AI tool.

- **Prohibited Uses:** AI should not be used to complete homework, quizzes, exams, or any other assessments outside the bounds explicitly permitted by the instructor and described here.

Consequences for Misuse: Misuse of AI tools in coursework, which includes but is not limited to producing unoriginal work, uncited use of AI-generated content, or unauthorized assistance on assessments, will be considered a breach of academic integrity. Consequences will follow the Purdue's policies on academic dishonesty.

Example of appropriate use: While processing data for a ME315 lab report, you ask ChatGPT to help you write a template Matlab code for a function you haven't use before; you check the original Matlab documentation to understand the function, test the accuracy of the code for a sample data set, and clearly cite the use of ChatGPT as a code-writing collaborator for this specific task performed in the report.

Example of misuse: You ask ChatGPT to answer several of the lab report discussion questions, copy/paste them into your lab report without considering or confirming their validity and fail to find an original source of information that led ChatGPT to the generated response.

Example of appropriate use: After completing a homework problem, you describe your approach and ask Gemini if the methodology and assumptions are reasonable. It replies that you have made an oversimplification and that a more complex analysis was needed. You cross-check the AI response with your lecture notes and textbook concluding that Gemini was correct. As such, you revise your homework submission with the new methodology including a description of how Gemini helped you find your mistake.

Example of misuse: You submit a homework problem prompt to Gemini and utilize the solution provided from AI modifying it only slightly to "put it in your own words." The solution was generated by AI but enhanced by you rather than vice versa. The logic and methodology were machine—instead of student—generated.

Example of appropriate use: You are unsure how to get started on a homework problem set because you found the concepts covered in lecture difficult to understand (or missed a lecture). After review of the associated textbook sections, you identify the specific concept that you did not understand and ask Copilot to provide you an explanation of this concept in 5 levels of difficulty, along with short example problems at the college-level difficulty. After reading these outputs, you are then able to understand the solutions to the example problems in the textbook and further verify the AI-generated example solutions. You acknowledge this use of Copilot in your homework solution as an AI tutor that helped you learn the concept.

ME 315: Heat and Mass Transfer
Class and Assignment Schedule

Text: T.L. Bergman, A.S. Lavine, *Fundamentals of Heat and Mass Transfer*, Eighth Edition, John Wiley. Other editions are okay but please note the section and problem numbers may be different. All homework statements will be posted to the course-wide Gradescope and Brightspace pages.

Maps to	Week	Meeting		Date	Topics	Book Sections	HW Content	Lec. HW
Exam #1 Conduction Phenomena	1	1	M	12-Jan	Introduction, Rate Equations	1.1, 1.2.1, 1.2.2	HW #1	HW#1 Post
		2	W	14-Jan	Rate Equations, Conservation laws	1.2.3-1.3.1	HW #1	
		3	F	16-Jan	Introduction to Conduction	2.1, 2.2	HW #1	
	2		M	19-Jan	**NO CLASS Martin Luther King Day**			HW#2 Post
		4	W	21-Jan	Conduction Conservation Equations	2.3-2.5	HW#2	**HW#1 Due**
		5	F	23-Jan	1D Steady State Conduction, Thermal Resistance	3.1.1-3.1.4, 3.2	HW#2	
	3	6	M	26-Jan	Variable Area, Radial Conduction	3.3,3.4	HW#3	HW#3 Post
		7	W	28-Jan	Heat Generation	3.5	HW#3	**HW#2 Due**
		8	F	30-Jan	Extended Surfaces (i.e., Fins)	3.6-3.6.2	HW#3	
	4	9	M	2-Feb	More Extended Surfaces	3.6.2-3.7	HW#4	HW#4 & Ex. 1 Formula Post
		10	W	4-Feb	2D Steady State Conduction	4.1-4.3	HW#4	**HW#3 Due**
		11	F	6-Feb	Transient Conduction-Lumped Capacitance	5.1-5.3	HW#4	
	5	12	M	9-Feb	Transient Conduction (Non-Lumped)	5.4-5.7	HW#5	HW#5 Post
		13	W	11-Feb	Finite Difference Methods (Steady State)	4.4-4.5	HW#5	**HW#4 Due**
		14	F	13-Feb	Finite Difference Methods (Transient)	5.10	HW#5	
	6	15	M	16-Feb	Review for Exam #1 (Fall 2025 Exam Review)			HW#5 is
		16	**W**	**18-Feb**	**Exam #1 8:00PM-9:30PM**			**Not Graded**
Exam #2 Convection Phenomena	6	17	F	20-Feb	The Problem of Convective Transport	6.1-6.3	HW#6	
	7	18	M	23-Feb	Convective Transfer Equations and Similarity	6.4-6.6	HW#6	HW#6 Post
		19	W	25-Feb	Heat and Mass Transfer Analogy	6.7-6.7.2	HW#6	
		20	F	27-Feb	External Flow (Flat Plate)	7.1-7.3	HW#6	
	8	21	M	2-Mar	External Flow (Other Geometries)	7.4-7.5	HW#7	HW#7 Post
		22	W	4-Mar	Internal Flow Intro	8.1-8.2	HW#7	**HW#6 Due**
		23	F	6-Mar	Internal Flow Applications	8.3-8.5	HW#7	
	9	24	M	9-Mar	Free Convection Intro	9.1-9.3	HW#8	HW#8 & Ex. 2 Formula Post
		25	W	11-Mar	Free Convection Applications	9.4-9.6	HW#8	**HW#7 Due**
		26	F	13-Mar	Boiling Heat Transfer	10.2-10.4	HW#8	
	10	27	M	16-Mar	**NO CLASS SPRING BREAK**			
		28	W	18-Mar	**NO CLASS SPRING BREAK**			
		29	F	20-Mar	**NO CLASS SPRING BREAK**			
	11	30	M	23-Mar	Condensation	10.6-10.7	HW#9	HW#9 Post
		31	W	25-Mar	Heat Exchangers: LMTD Method	11.1-11.2	HW#9	**HW#8 Due**
		32	F	27-Mar	Heat Exchangers: ε-NTU Method	11.3-11.5	HW#9	
	12	33	M	30-Mar	Review for Exam #2 (Fall 2025 Exam Review)			HW #9 is
		34	**Th**	**2-Apr**	**Exam #2 8:00PM-9:30PM**			**Not Graded**
Exam #3 Radiation + Cumulative	12	35	F	3-Apr	Thermal Radiation Concepts	12.1-12.2	HW#10	
	13	36	M	6-Apr	Radiation Intensity, Irradiation, Radiosity	12.3.1-12.3.2	HW#10	HW#10 Post
		37	W	9-Apr	Blackbody Emission, Real Surface Emission	12.4-12.5	HW#10	
		38	F	10-Apr	Absorption, Reflection, Transmission, Energy Balance	12.6	HW#10	
	14		M	13-Apr	Kirchoff's Law, Gray/Non-Gray	12.7-12.8	HW#11	HW#11 Post
			W	16-Apr	Environmental Radiation	12.9	HW#11	**HW#10 Due**
			F	17-Apr	The View Factor	13.1	HW#11	
	15	39	M	20-Apr	Blackbody Radiation Exchange	13.2	HW#12	HW#12 & Ex. 3 Formula Post
		40	W	23-Apr	Diffuse-Gray Surface Radiation Exchange	13.3-13.3.2	HW#12	**HW#11 Due**
		41	F	24-Apr	Enclosure Calculations	13.3.3-13.3.5	HW#12	
	16	42	M	27-Apr	Multimode Analysis	13.4	HW#12	**Quiet Week**
		43	W	30-Apr	Radiation All in One, Course Review			**No Quiz**
		44	F	1-May	Review for Exam #3 (Fall 2025 Exam Review)			**HW#12 Not Graded**
	17		M	4-May				
			W	7-May	**Final Exam (Date, Time and Location TBD)**			
			F	8-May				

HOMEWORK SOLUTION

EXAMPLE PAGE

Work problems on engineering paper in a handwritten manner. Note that "digitally handwritten" (i.e., written on a tablet) solutions are permissible. No more than one problem to a sheet. If a particular section shown below is irrelevant for a problem, it may be omitted.

KNOWN:
State concisely what is known about the problem.

FIND:
State concisely what must be found.

SCHEMATIC:
Draw a schematic of the physical system being considered. Label important variables. If application of the conservation laws is anticipated, represent the appropriate control volume or control surfaces by dashed lines. Be sure to identify processes associated with control volume/surfaces.

PROPERTIES:
List the solid and/or fluid thermophysical properties used in your solution. Identify the table from the Text and especially the temperature at which the property was selected.

ASSUMPTIONS:
It is important that you put all the assumptions in one place so that they can be reviewed. At the outset, some assumptions may be obvious, like "steady-state conditions," etc. But as you begin to model more complicated systems, the assumptions are extremely important to the logic of your analysis.

ANALYSIS:
Provide in sentence format, comments that make clear the logic and organization of your analysis. Be sure to identify by numbers any figures or equations taken from the Text.